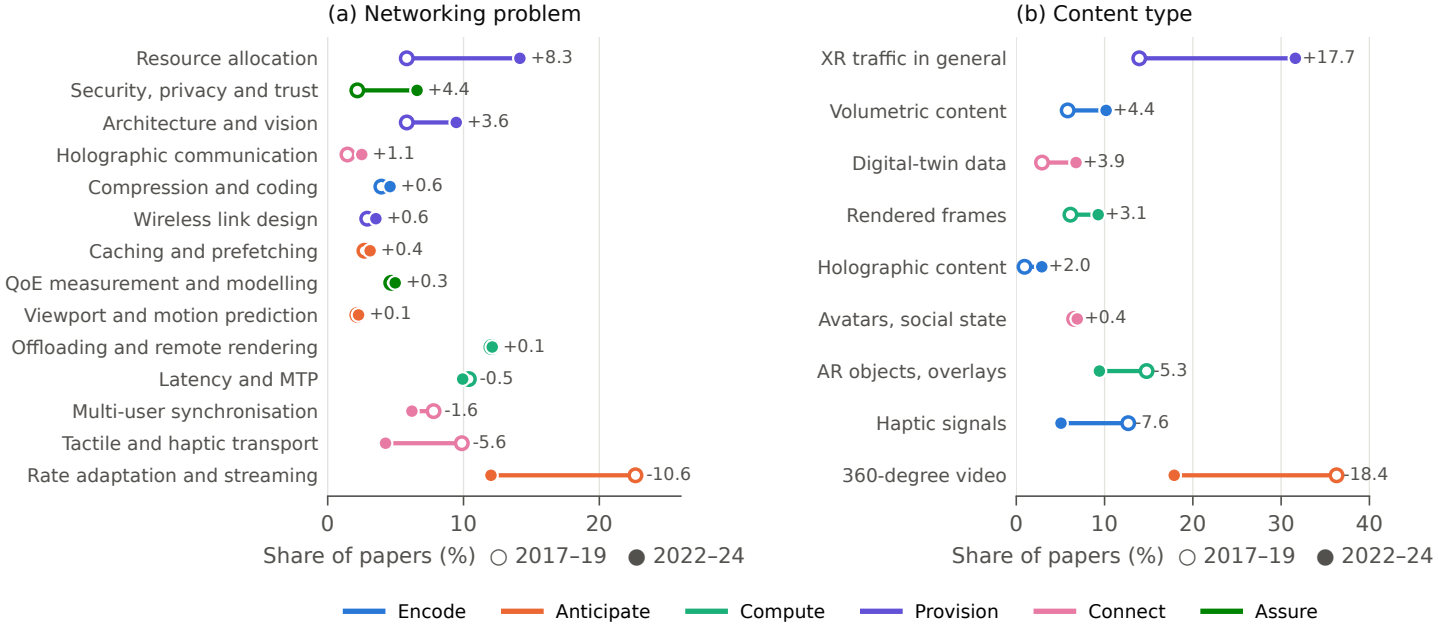
(a) Networking problem
Resource allocation
Security, privacy and trust
Architecture and vision
Holographic communication
Compression and coding
Wireless link design
Caching and prefetching
QoE measurement and modelling
Viewport and motion prediction
Offloading and remote rendering
Latency and MTP
Multi-user synchronisation
Tactile and haptic transport
Rate adaptation and streaming
+8.3
+4.4
+3.6
+1.1
+0.6
+0.6
+0.4
+0.3
+0.1
+0.1
-0.5
-1.6
-5.6
-10.6
0
10
20
Share of papers (%) ○ 2017–19 ● 2022–24
(b) Content type
XR traffic in general
Volumetric content
Digital-twin data
Rendered frames
Holographic content
Avatars, social state
AR objects, overlays
Haptic signals
360-degree video
+17.7
+4.4
+3.9
+3.1
+2.0
+0.4
-5.3
-7.6
-18.4
0
10
20
30
40
Share of papers (%) ○ 2017–19 ● 2022–24
Encode
Anticipate
Compute
Provision
Connect
Assure